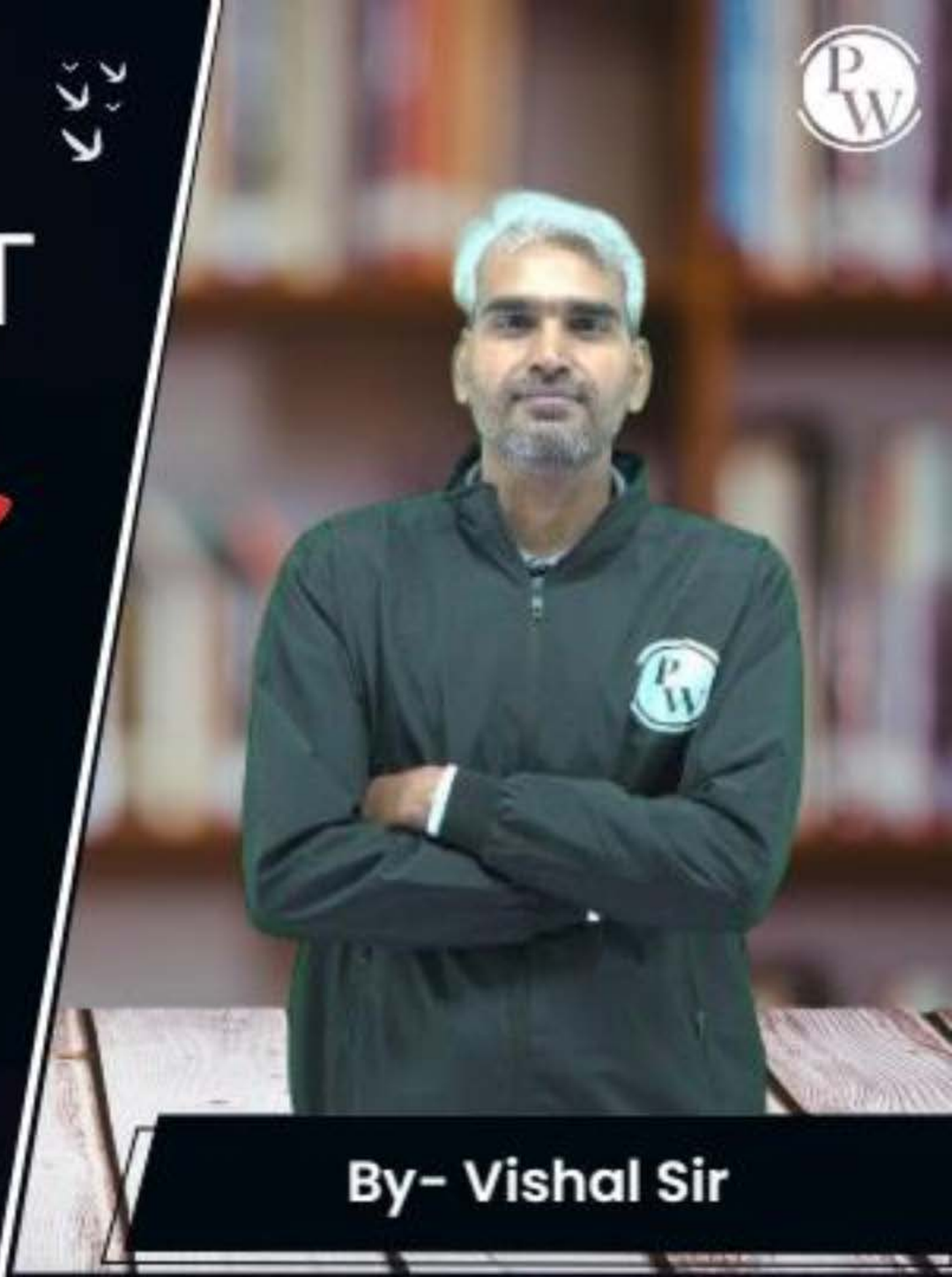
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 18



By- Vishal Sir

Recap of Previous Lecture



Topic — Distributive lattice

Topic — Boolean algebra

Slide

Topics to be Covered



- Topic: Functions , Domain and Codomain of a function
- Topic: One-one (Injective) & Onto (Surjective) functions
- Topic: Number of one-one functions
- Topic: Number of onto functions

Slide

H.W.

Q:1 Which of the following statements is/ are not true

* a) If A is any finite set then [P(A),⊆] is distributive lattice (True)
b) Every sub lattice of a distributive lattice is also a distributive lattice
c) Every totally ordered set is a distributive lattice (True)
d) Every totally ordered set is bounded False {(Z, ≤) is TOSET, but not bounded}
e) Every distributive lattice is bounded (Z, ≤) is distributive but not bounded (False)
f) Every distributive lattice is a complemented lattice False (D12, |)

Let [M, ∨, ∧] be a sub-lattice of lattice [L, ∨, ∧]

∀ a, b, c ∈ M we know a, b, c ∈ L as well

We already know L is distributive

i.e. $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

& $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

We know lub & glb in sub-lattice for any pair of elements will remain same as original lattice.

∴ Same distributive property will hold true in sub-lattice [M, ∨, ∧] as well.

Hence [M, ∨, ∧] will also be distributive

Slide

H.W.

Which of the following lattice is /are not distributive?

(A) $(D_{125}, \div)$ $(\{1, 5, 25, 125\}, \div)$ ⇒ Size of lattice = 4 ∴ distributive

(B) $(P(A), \subseteq)$ is distributive

(C) $(D_{12}, \div)$ is distributive

(D) $(\{1, 2, 3, 5, 30\}, \div)$ is isomorphic to L_2^*, ∴ Not distributive

(E) $(\{1, 2, 3, 4, 12\}, \div)$ is isomorphic to L_1^* ∴ Not distributive

Slide

H.W.

Which of the following lattice is /are not Complemented ?

(A) $(D_{125}, \div)$ TOSET with more than '2' elements, ∴ Not Complemented

(B) $(P(A), \subseteq)$ Every element have unique Complement ∴ Complemented

(C) $(D_{12}, \div)$ $\bar{2}$ = DNE & $\bar{6}$ = DNE ∴ Not Complemented

(D) $(\{1, 2, 3, 5, 30\}, \div)$ } Both are Complemented

(E) $(\{1, 2, 3, 4, 12\}, \div)$ }

Slide

H.W.

Which of the following lattice is /are boolean lattice ?

(A) $(D_{125}, \div)$ Distributive, but not Complemented

(B) $(P(A), \subseteq)$ Distributive as well as Complemented

(C) $(D_{12}, \div)$ Distributive but not complemented

(D) $(\{1, 2, 3, 5, 30\}, \div)$ } Complemented but not distributive

(E) $(\{1, 2, 3, 4, 12\}, \div)$ }

Slide

Functions

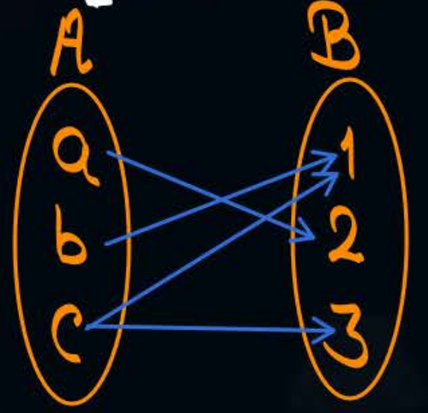


Topic : Function

Every function from set A to set B is also a relation from A to B, but Every relation from set A to set B need not be a function from A to B

- A relation from set A to set B is called a function from set A to set B if and only if Every element of set A relates with exactly '1' element of set B.

eg: $R_1 : A \to B$ { R_1 is a relation from A to B }



{ element 'c' of set A relates with two elements of set B, ∴ Not a function }

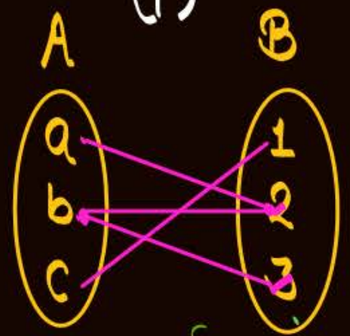
eg: $R_2 : A \to B$ { R_2 is a relation from A to B }



{ Element 'b' relates with '0' element of set B, ∴ Not a function }

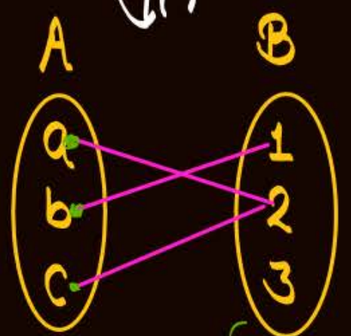
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(i)



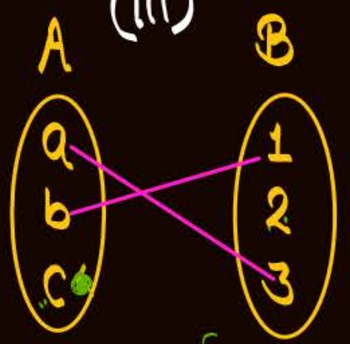
Neither a function from A to B, nor a function from B to A

(ii)



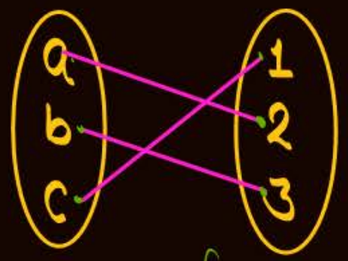
It is a function from A to B, but not a function from B to A

(iii)



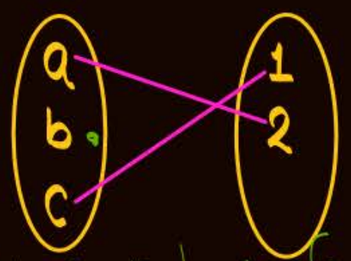
Neither a function from A to B, nor a function from B to A

(iv)



It is a function from A to B as well as a function from B to A

(v)



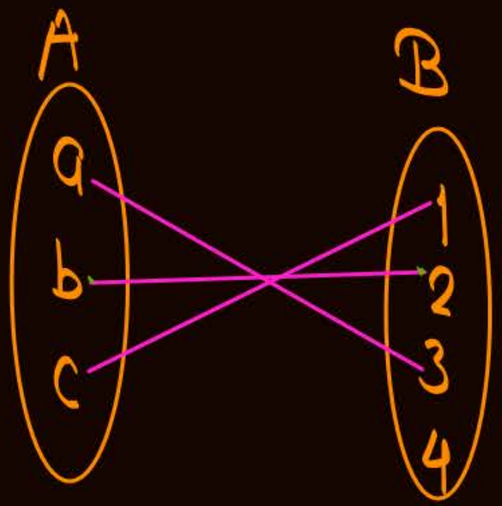
It is not a function from A to B, but a function from B to A

- Let $f: A \to B$ is a function,

then set 'A' is called domain of the function 'f'

and set 'B' is called Co-domain of the function 'f'

Eg: $f: A \to B$



domain of f = {a, b, c}

Co-domain of f = {1, 2, 3, 4}

Here $f(b) = 2$,

i.e. image of element 'b' w.r.t. function 'f' is '2'.

(or)

We say that Pre-image of '2' w.r.t. function 'f' is 'b'.



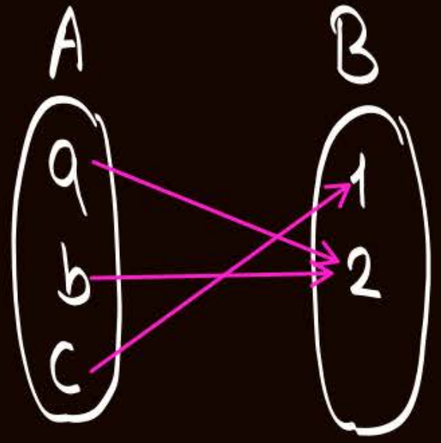
Topic : Range of a function

The set of all elements from Co-domain that are mapped by at least one element of domain is called "Range" of the function.

Range of function $f \subseteq$ Co-domain of function f

Slide

eg: $f_1 : A \to B$



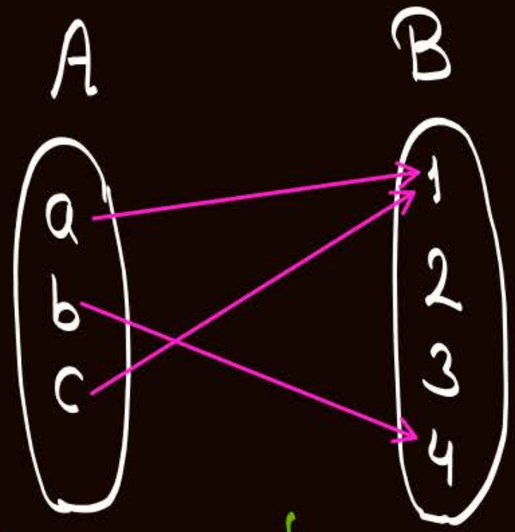
Domain : {a, b, c}

Co-domain : {1, 2}

Range = {1, 2}

Range = Co-domain

$f_2 : A \longrightarrow B$



Domain : {a, b, c}

Co-domain = {1, 2, 3, 4}

Range = {1, 4}

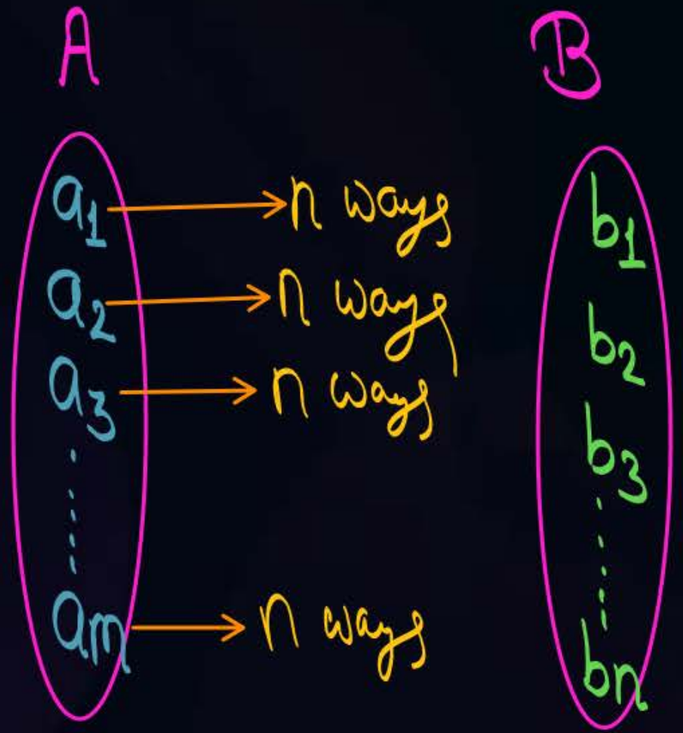
Range $\subset$ Co-domain



Topic : Total number of functions

* Let $|A| = m$ and $|B| = n$, then

Total Number of Functions Possible from Set A to Set B $= n^{m}$ (Size of Codomain ^ Size of domain)



∴ Total no. of functions Possible from Set A to Set B $= \underbrace{n * n * \cdots * n}_{m \text{ times } n}$

$= n^{m}$

Slide

Note:- A function defined from set A to set A itself is called a function on set A.

If $|A| = n$, then

Total no. of functions possible on set A (from A to A) $= (\text{Size of Codomain})^{\text{Size of domain}} = (|A|)^{|A|} = n^n$



Topic : Injective (one-one) function

A function $f: A \to B$ is called an injective (one-one) function if and only if:

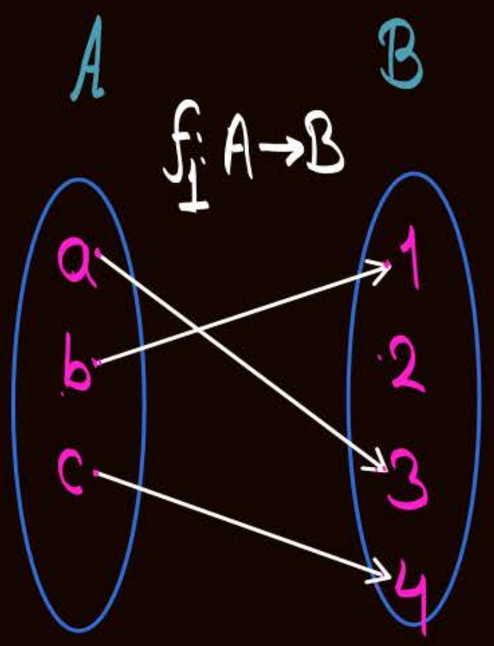
* distinct elements of the domain (set A) have distinct images in codomain (set B)

i.e. $\forall\, a, b \in$ domain

if $a \neq b$ then $f(a) \neq f(b)$

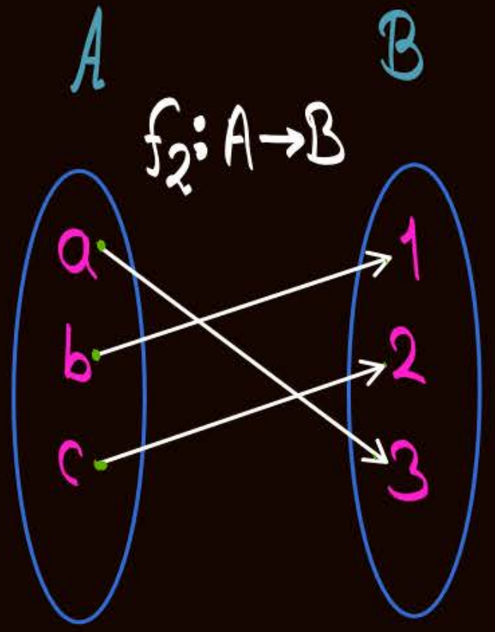
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$|A| < |B|$



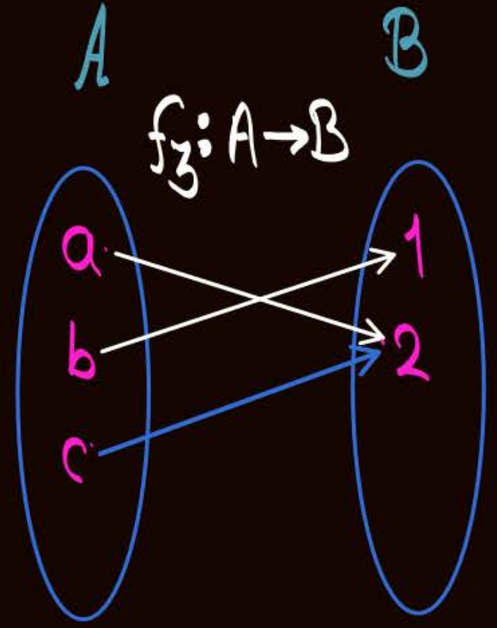
'f_1' is a One-one function from A to B

$|A| = |B|$



'f_1' is a One-one function from A to B

$|A| > |B|$



$a \neq c$ but $f(a) = f(c)$

$\therefore$ Not a one-one function

'f_3' is not one

Note:-

① If $|A| > |B|$, then
One-one function is not possible from set A to set B

② If One-one function exists from set A to set B, then $|A| \leq |B|$

③ In one-one function Every element of Co-domain can have at most '1' pre-image

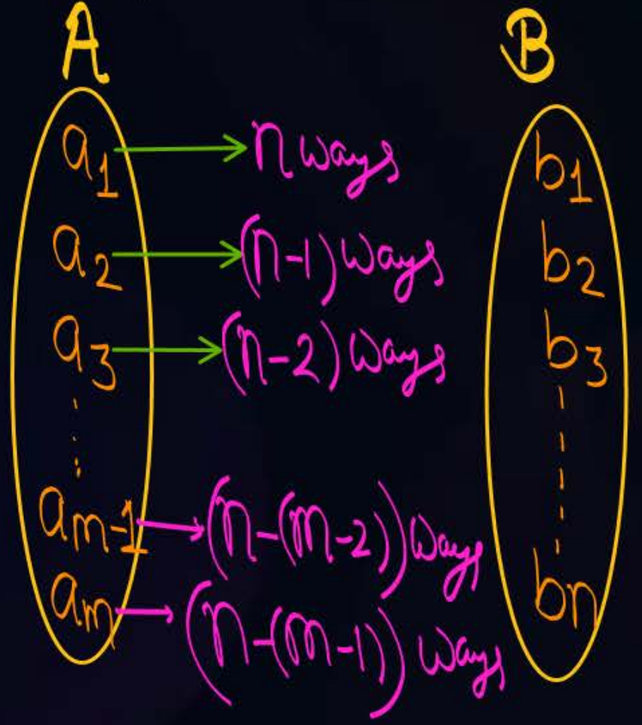


Topic : Number of one-one function

Let $|A| = m$ & $|B| = n$ (such that $m \leq n$), then,

Number of one-one functions possible from set A to set B $= \frac{n!}{(n-m)!} = {}^{n}P_{m}$

$\because {}^{n}P_{r} = \frac{n!}{(n-r)!}$



Number of one-one functions possible from set A to set B $= n*(n-1)\times(n-2)\times\cdots*(n-m+2)(n-m+1)$

$= n(n-1)(n-2)\ldots(n-m+1) * \frac{(n-m)(n-m-1)\ldots3.2.1}{(n-m)(n-m-1)\ldots3.2.1}$

$= \frac{n!}{(n-m)!}$

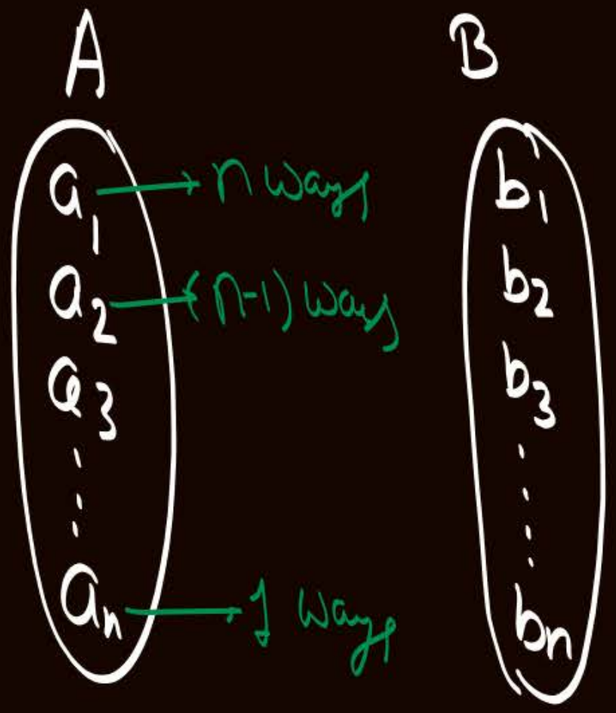
$\therefore$

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Note: If $|A| = |B| = n$,

then, number of one-one function Possible from set A to set B $= {}^{n}P_{n} = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$

$0! = 1$



No. of one-one function Possible $= n(n-1)(n-2)\ldots 3 \cdot 2 \cdot 1$

$= n!$



Topic : Surjective (onto) function

A function $f: A \to B$ is called an Onto (Surjective) function if and only if

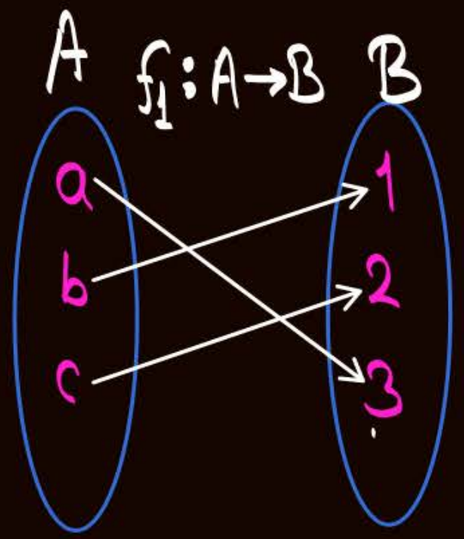
Every element of Co-domain is mapped by at least one element of domain

i.e. in onto function.

Range = Co-domain

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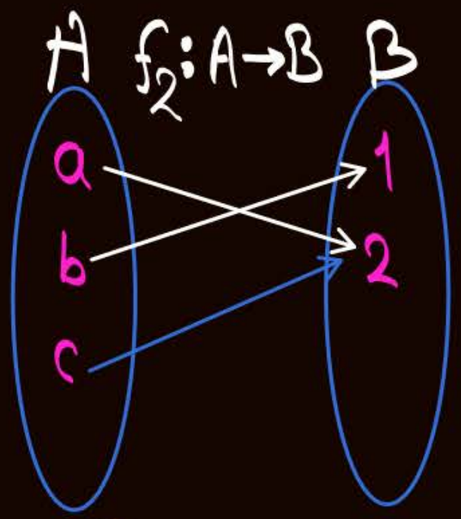
$|A| = |B|$



Range = Co-domain

$\therefore f_1$ is an onto function

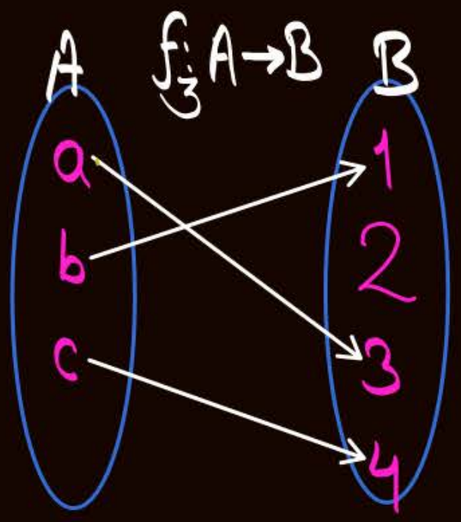
$|A| > |B|$



Range = Co-domain

$\therefore f_2$ is an onto function

$|A| < |B|$



Range $\subset$ Co-domain

$\therefore f_3$ is not an onto function

Note:

① If $|A| < |B|$, then Onto function is not possible from set A to set B.

② If there exist an onto function from set A to set B, then $|A| \geq |B|$

③ In an onto function Every element of Co-domain will have at least '1' pre-image



Topic : Number of onto functions

① If $|A| = |B| = n$, then
How many Onto functions are possible from A to B

② If $|A| = n$ & $|B| = 2$ such that $n \geq 2$, then
How many onto functions are possible from A to B

③ if $|A| = 4$ & $|B| = 3$, then
How many onto functions are possible from A to B.

Slide

H.W.

Q:- If $|A| = |B| = n$
then How many onto functions are possible from A to B.

H.W.

Q:- If $|A|=n$ and $|B|=2$ $(s.t.\ n \geq 2)$ then How many onto functions are possible from A to B.

H.W. Q:- If $|A|=4$ and $|B|=3$ then How many onto functions are possible from A to B.

H.W.

Q:- If $|A| = n$ and $|B| = n-1$
then How many onto functions are possible from A to B.

H.W.

Q: if $|A| = m$ & $|B| = n$ (such that $m \geq n$)
then how many onto functions are possible from set A to set B.



2 mins Summary

Topic Functions , Domain and Codomain of a function

Topic One-one (Injective) & Onto (Surjective) functions

Topic Number of one-one functions

Topic Number of onto functions

Slide

THANK - YOU